UNIVERSITI TEKNOLOGI MARA
REMEDIAL TEST

COURSE	:	ELECTRONIC CIRCUIT ANALYSIS
COURSE CODE	:	ETL512
DATE	:	FEB 2026
TIME	:	45 MINUTES

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of **ONE (1)** questions The total marks is 15.
2. Answer ALL questions.
3. Please attach this cover page together with your answer script.

FILL IN THE FOLLOWING PARTICULARS
NAME: Muhammad Daniel Hakim bin Zamzuhari
GROUP: PEE2113A2 GRADED MARKS: _____/15

FILL IN THE FOLLOWING PARTICULARS (FOR OFFICE)

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COURSE	ETL512	TEST	1						
Question	CO	PO	Allocated Marks	Graded Marks	Question	CO	PO	Allocated Marks	Graded Marks
1	CO2	PO2	15						

OBE IMPLEMENTATION

BLOOM'S LEVEL FOR SEM 3		Level I (35%)	Level II (65%)
BLOOM'S LEVEL FOR TEST 1	QUESTION		Q1
	DOMAIN COGNITIF		
	COURSE OUTCOME		CO2
	PROGRAM OUTCOME		PO2
ALLOCATED MARKS			15

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QUESTION 1

Figure shows a BJT amplifier circuit. The amplifier parameters are, V_T = 26 mV, β = 100, I_E = 5.7 mA, r_{bb}' = 10 Ω, and r_{ce} = 56 kΩ.

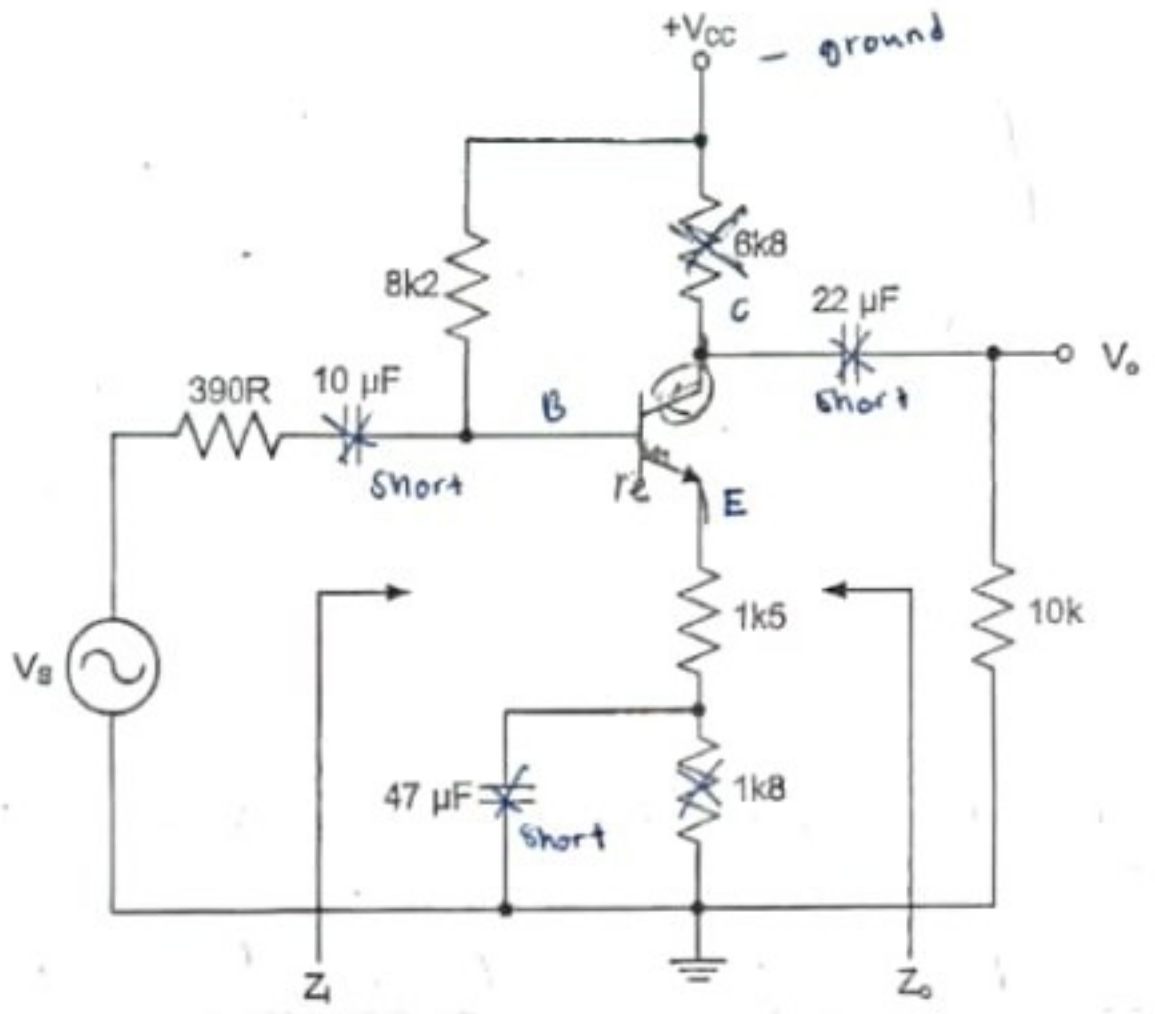


Figure Q1

i) Illustrate AC equivalent circuit.
ii) Determine dynamic resistance, r_e.
iii) Calculate the input impedance, Z_i.
iv) Calculate the output impedance, Z_O.
v) Calculate the voltage gain, A_V.
vi) Calculate current gain, A_i.

(15 marks)

END OF QUESTION PAPER

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Subject :

Date:

Question 1

Given

V_T = 26mV

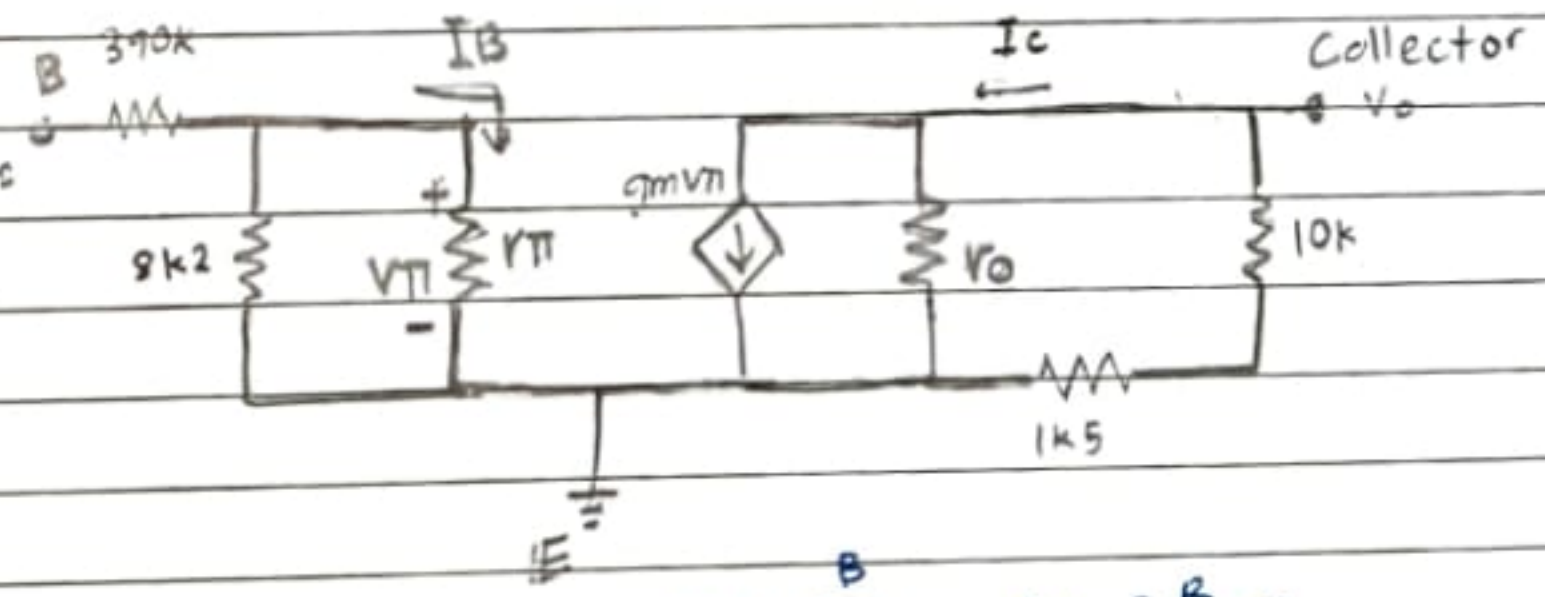
$\beta = 100$

$I_E = 5.7mA$

$r_{bb'} = 10\Omega$

$r_{ce} = 56k\Omega$

i) AC equivalent circuit



$I_C = I_E$

$I_E = 5.7mA$

V_{CE} V_C

g_m

$r\pi$

V_o

$I_C = 5.7mA$

$g_m = \frac{I_C}{V_T}$

$= \frac{5.7mA}{26mV} = \frac{0.0057A}{0.026V}$

$= 0.219$ Siemens

ii) $r_e = \frac{V_T}{I_E} = \frac{26mV}{5.7mA}$

$= 4561$ Ohm 4.56Ω

$r\pi = \frac{\beta}{g_m}$ $r_\pi = \beta . re$

$= \frac{100}{0.219}$

$= 456\Omega$

$V\pi = I_b \times R\pi$

$= 57\mu A \times 456\Omega$

$= 0.025992\ V$

$= 25\ mV$

$\beta = \frac{I_c}{I_b}$

$100 = \frac{5.7mA}{I_b}$

$I_B = \frac{5.7mA}{100}$

$I_b = 57\mu A$

iii) Input Impedance Z_i

$Z_i = 390R + (8k2 // r\pi)$

$= 390R + (8k2 // 456\Omega) = 390R + 432R = 822\Omega$ ✱

Subject : Date:

iv) Output Impedance Z_o

$Z_o = 10k + 1k5$

$= 11.5k\Omega$

v) Voltage gain Av

$$Av = \frac{V_{out}}{V_{in}} = -g_m (R_C // r_o)$$

$= -0.219\ (z_o)$

$= -0.219 \times 11.5k\Omega$

$= -2518$ ✗

vi) Current gain Ai

$$Ai = \frac{I_{out}}{I_{in}}$$

$$= \frac{I_c}{I_b}$$

$$= \frac{5.7mA}{57\mu A}$$

$$= \frac{0.0057}{0.000057A}$$

$= 100$ ✗